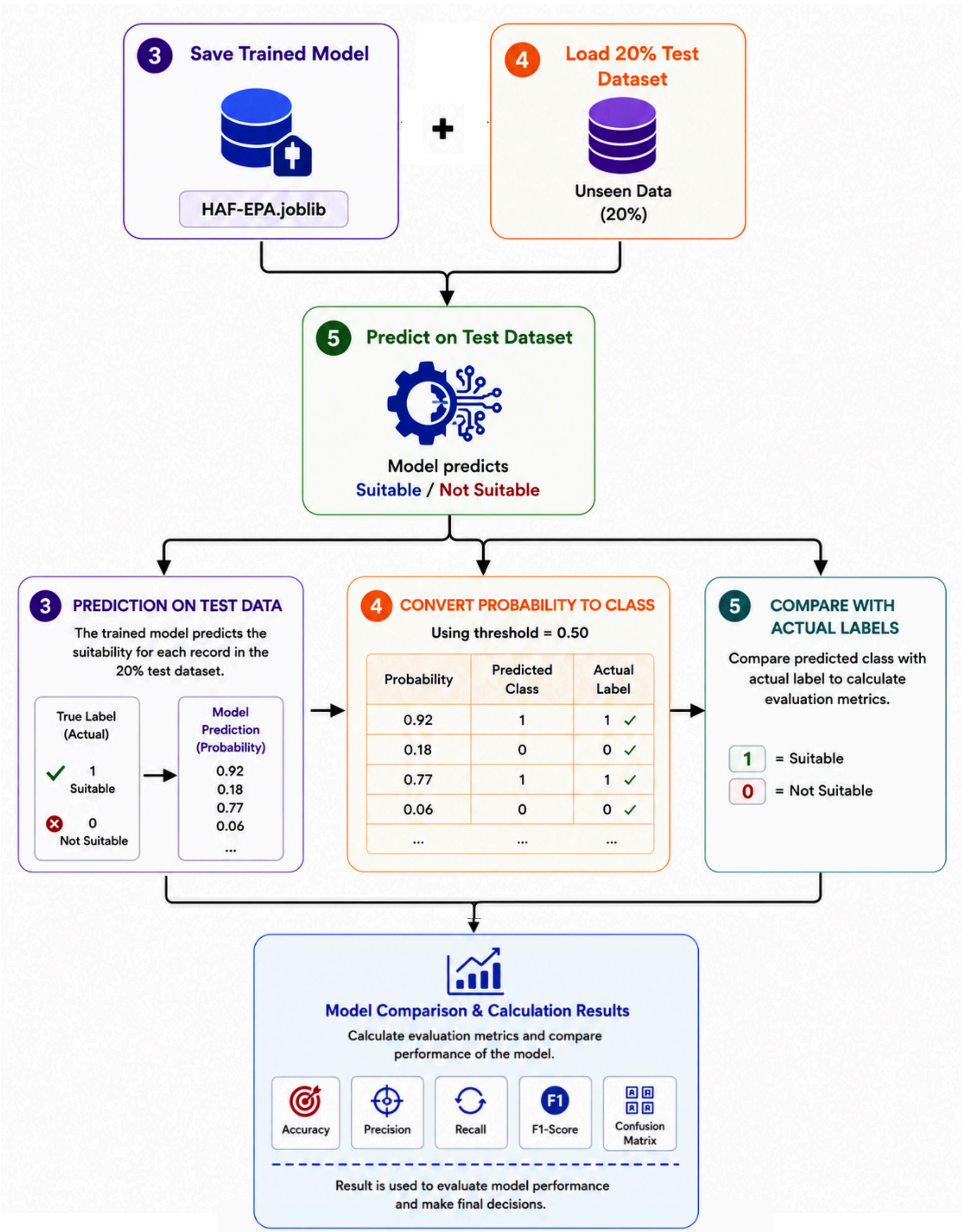
3 Save Trained Model
HAF-EPA.joblib
4 Load 20% Test Dataset
Unseen Data (20%)
5 Predict on Test Dataset
Model predicts
Suitable / Not Suitable
3 PREDICTION ON TEST DATA
The trained model predicts the suitability for each record in the 20% test dataset.
True Label (Actual)
1 Suitable
0 Not Suitable
Model Prediction (Probability)
0.92
0.18
0.77
0.06
...
4 CONVERT PROBABILITY TO CLASS
Using threshold = 0.50
Probability | Predicted Class | Actual Label
0.92 | 1 | 1
0.18 | 0 | 0
0.77 | 1 | 1
0.06 | 0 | 0
... | ... | ...
5 COMPARE WITH ACTUAL LABELS
Compare predicted class with actual label to calculate evaluation metrics.
1 = Suitable
0 = Not Suitable
Model Comparison & Calculation Results
Calculate evaluation metrics and compare performance of the model.
Accuracy
Precision
Recall
F1
F1-Score
Confusion Matrix
Result is used to evaluate model performance and make final decisions.

Model Evaluation on Test Dataset (20%)

After training is completed, the saved **Random Forest model (HAF-EPA.joblib)** is evaluated using the **20% unseen test dataset** to measure its real-world performance.

Step 1: Load the Trained Model
The previously trained and saved model (HAF-EPA.joblib) is loaded for evaluation.

Step 2: Load the Test Dataset
The remaining 20% of the dataset, which was not used during training or cross-validation, is loaded as unseen test data.

Step 3: Generate Predictions
The model predicts the suitability of each employee-project pair in the test dataset.
For each record, the model outputs:

- A probability score (0–1)
- A predicted class:
  - **1 = Suitable**
  - **0 = Not Suitable**

Step 4: Convert Probabilities to Classes

Using a decision threshold of **0.50**:

- Probability ≥ 0.50 → Suitable (1)
- Probability < 0.50 → Not Suitable (0)

Step 5: Compare with Actual Labels
The predicted results are compared with the actual labels from the test dataset to determine how accurately the model performs on unseen data.

Step 6: Calculate Evaluation Metrics
Several performance metrics are calculated:

- **Accuracy** – Overall prediction correctness.
- **Precision** – How many predicted suitable employees are actually suitable.
- **Recall** – How many truly suitable employees are successfully identified.
- **F1-Score** – Balanced measure of precision and recall.
- **Confusion Matrix** – Detailed comparison of predicted and actual classes.

Outcome

The evaluation results provide a reliable assessment of the model's ability to recommend suitable employees for projects. Since the test dataset was never seen during training, these metrics represent the model's expected performance in real-world deployment.